Sādhanā (2022)47:111
https://doi.org/10.1007/s12046-022-01883-6

Effect of particle doping on the mechanical behavior of 2D woven (0°/90°) jute fabric (plain weave) reinforced polymer matrix composites

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MS received 8 May 2021; revised 18 October 2021; accepted 8 April 2022

Abstract. The performance and functionality of natural fiber reinforced polymer composites can be enhanced many folds by incorporating hard filler particulates which may significantly increase mechanical properties. This study was aimed to hybridize the jute fabric reinforced polymer (JFRP) based laminated composites with cement particulates for the enhancement of their mechanical properties. The laminated cement particulates filled jute fabric reinforced polymer (CmJFRP) and JFRP composites were fabricated via hand lay-up method. The weight fraction of cement particles was varied as 1%, 3%, and 5%. The thickness of the composites was also varied as 3.1 mm, 4.4 mm, and 6.3 mm (i.e. by varying the number of lamina). The mechanical behavior of the composites was assessed via tensile test, microhardness, Izod impact test, and critical buckling load. The tensile tests revealed that CmJFRP composites exhibit lower tensile strength than that of JFRP composites, which was mainly ascribed to the agglomeration of cement particles at the fiber/matrix interface. The microhardness, impact strength, and critical buckling load were higher for CmJFRP composites as compared to JFRP composites. The scanning electron microscopy of the fractured composites revealed that crack front changed its plane and direction when it encountered the cement particles.

Keywords. Jute fibers; polymer; cement particles; tensile strength; impact test; critical buckling load.

1. Introduction

Recent years have witnessed a significant pace in the research of fiber reinforced polymer matrix composites (FRPs) due to their appealing properties such as high specific strength, resilience, corrosion resistance, creep resistance, and excellent damping properties [1]. The FRPs are being used in a variety of applications like marine, automotive, aerospace, anti-vibration and structural. The properties of FRPs depend on the type of matrix, fiber, and fiber/matrix interface [2]. It is relatively easy to choose among various types of matrix and fiber materials, but it is difficult to engineer the fiber/matrix (F/M) interface. The properties of F/M interface need to be understood to tailor the properties of FRPs. Some researchers [3–6] have successfully enhanced the adhesion of fiber with matrix by pre-functionalizing the fibers with suitable molecules, which improves the interfacial properties. However, functionalizing of fibers require specialized set-ups and significant knowledge of functionalizing mechanisms at the molecular level. The other method includes loading of FRPs with suitable particulate fillers. This method may or may not enhance the interfacial properties of F/M interface, but usually improves the properties of matrix which results in improvement of the properties of the composites [7]. Although synthetic fibers exhibit excellent properties, they are highly expensive. Thus, synthetic fibers are mainly limited to military and aircraft applications. Now-a-days, natural fibers have gained significant attention of scientists and researchers for potential reinforcement in FRPs for their use mainly in civil structures and consumer goods. Natural fibers have several advantages over synthetic fibers such as low cost, low density, nontoxicity, biodegradability, combustibility, renewability, and high specific mechanical properties [8]. But natural fibers possess low strength as compared to synthetic fibers. The mechanical properties of natural fibers reinforced polymer composites could be enhanced by the addition of particulate fillers. Several particulates have been loaded in natural fibers reinforced polymer composites [9] such as red mud [10], copper slag

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Published online: 10 June 2022

particles [11], coir shell particles [12], ethylene acrylate [13], and oil palm [14], which improved the properties of matrix, and thereby, composites. Thus particulates filled natural fibers reinforced polymer matrix composites may exhibit better properties as compared to their unfilled counterparts.

Among various natural fibers, jute is most commonly used natural fiber due to its very low cost and its abundance growth in several countries such as Thailand, Nepal, India, Indonesia, China, Bangladesh, and Brazil [15]. It is a type of bast fiber, and belong to Tiliaceae family. Although jute has less resistance to UV light, acid and moisture, its resistance to heat and fire, and its fine texture makes it demanding in several industries such as automotive, construction and textile [16]. The mechanical properties of jute fiber reinforced polymer composites could be enhanced by loading the composites with particulate fillers [17–20]. Although cement particles enhances the bonding in concrete [21], and mechanical properties in glass fiber reinforced polymers [22], its effect on the natural fibers reinforced polymer matrix composites has not been investigated yet. Now-a-days the JFRP composites are considered as alternatives for several mechanical and wear applications due to their low weight and product cost, and enhanced renewability, but their performance is still low [23, 24]. The reinforcement of cement particles may exhibit positive influence on the mechanical properties of JFRP composites. Thus, the present article focuses on the hybridization of JFRP composites with cement particles and detailed investigations on the mechanical properties of cement particles loaded jute fibers reinforced polymer matrix composites.

2. Materials and methods

2.1 *Materials*

The laminated composites used in the present investigation were fabricated via hand lay-up technique. The composites consisted of cement particles filled, and pristine 2D woven (0°/90°) jute fabric (plain weave, 48 yarns in weft direction and 57 yarns in warp direction per 100 mm) reinforced in epoxy resin. The jute fabric was procured from Ved Fab, Varanasi, India, and the weight of jute fabric was 198 g m^{-2}. To fabricate pristine laminated composites, a weighed quantity of epoxy resin (PL-411) was taken in a container, and hardener (PH-861) was thoroughly mixed in it in the ratio of 1:10 by weight. The mixing was performed for 10 min, using a mechanical stirrer having 160 RPM. The jute fabric reinforced laminated composites (JFRP) were then prepared by hand lay-up technique. The composite plates were left for curing for 18 h at room temperature. But before that, a pressure of 8 KPa was applied on the un-cured composite for the proper infiltration of epoxy resin between the layers of jute. Same procedure was followed to fabricate cement filled jute fabric reinforced laminated composites (CmJFRP) with only difference that cement particles were mixed with epoxy resin prior to mixing with hardener. The weight percentage of cement particles was also varied as 1%, 3%, and 5% by weight of epoxy. The number of plies of jute fabric were also varied to obtain the thickness of composites as 3.1 mm (5 plies, slenderness ratio = 268.189), 4.4 mm (8 plies, slenderness ratio = 188.950), and 6.3 mm (11 plies, slenderness ratio = 131.966). The volume fraction of the fibers approximately varied from 0.37 to 0.43, considering all the fabricated samples. The JFRP composites can be potentially used for several mechanical applications [24], therefore samples were prepared for tensile test, impact test, hardness, and buckling tests, after curing of composite plates. All the samples were polished with emery paper before testing.

2.2 *Mechanical testing*

The samples were prepared from composite plates. For tensile test, ASTM D3039 standard was followed. For Izod impact testing, ASTM D256 was adopted. The hardness tests were performed on Vickers microhardness tester. The buckling test was performed on the Compression test bench. An external dial gauge was used to measure the mid-height lateral deflection of the specimen. The load was measured using a dial indicator set above the upper head, as shown in figure 1. The ends of the samples were free while performing buckling tests.

3. Results and discussion

Figure 2 shows variation of tensile strength of composites with variation of weight percentage of filler and thickness of composites.

It can be observed that tensile strength increased with increase in thickness of the composites. This was mainly ascribed to the delay of crack initiation due to thickness effect [25]. The increase in content of cement particles resulted in decrease in tensile strength. The agglomerates of cement particles were adhered on the surface of jute fibers (shown later in figure 9). These agglomerates acted as crack initiation sites when tensile stress was applied, and hence, resulted in decrease in strength with increase in percentage of cement particles. Furthermore, the decline in tensile strength may also be ascribed to the irregular shape of cement particles which acted as stress concentration sites.

The variation of hardness with the content of cement particles is shown in figure 3 (Note. The representative plot of hardness is presented for the composite having 6.3 mm thickness).

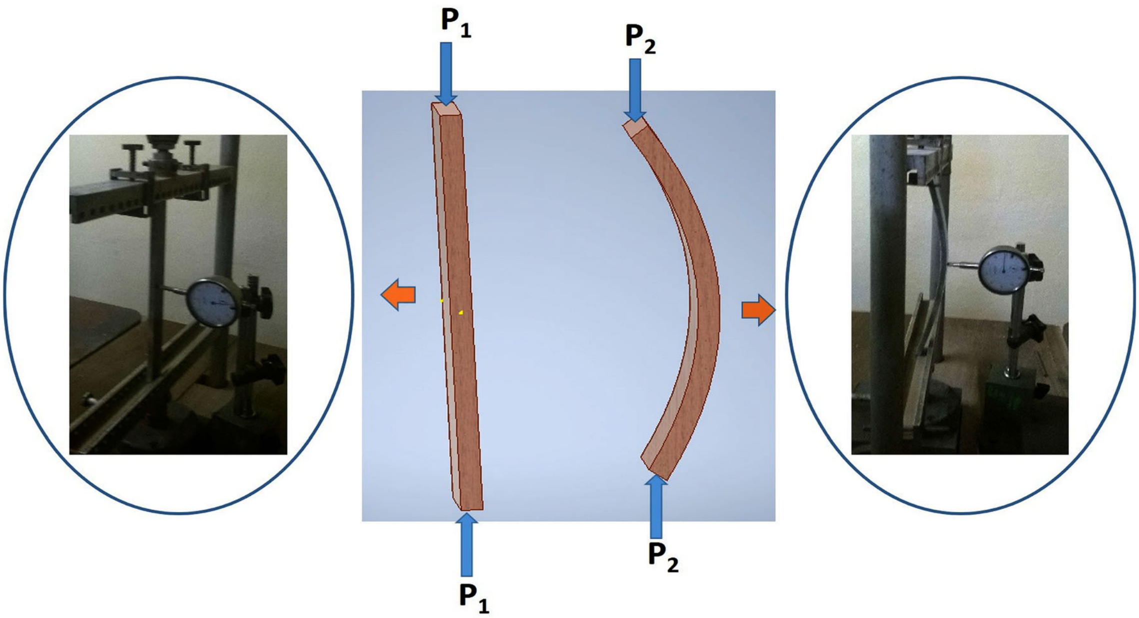

Figure 1. Image showing schematics and corresponding real photographs of buckling test.

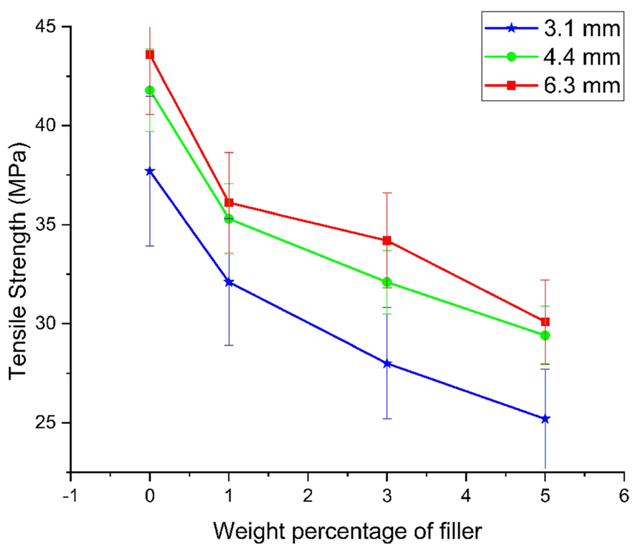

Figure 2. Variation of tensile strength with percentage of cement particles and thickness of composites.

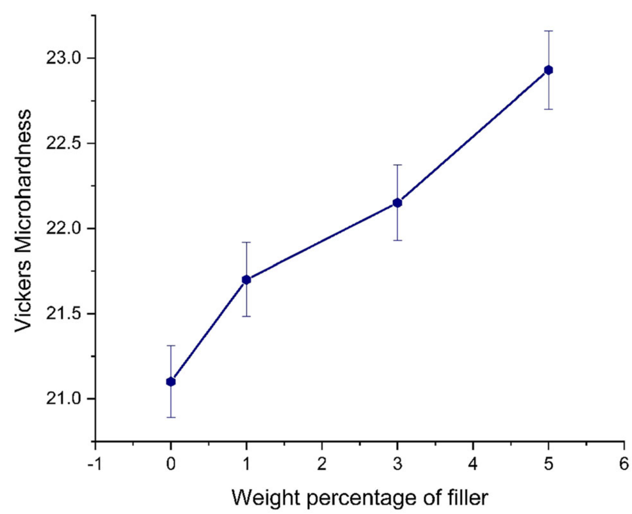

Figure 3. Variation of micro hardness with percentage of cement particles (plotted for composite having 6.3 mm thickness).

It can be observed from figure 3 that hardness increased with increase in the percentage of cement particles. The increase in hardness may be attributed to the increase in the percentage of hard cement particles. The compressive forces generated at the time of hardness test tries to press the filler and matrix close to each other [26]. The hard cement particles resisted the deformation which resulted in increased hardness of the composites.

The Izod impact resistance (i.e., impact energy absorbed up to fracture) increased with increase in the weight content of cement particles, as shown in figure 4.

If the adhesion at the interface of filler and matrix is not strong, the brittleness of ductile matrices is increased with the addition of particulate fillers. However, the reverse is true when brittle matrices are used [27]. Figure 5 shows the fractured surface of CmJFRP composite with 3% cement content and 3.1 mm thickness.

It can be observed that the direction and plane of crack front were changed at some locations when it encountered the cement particles. Thus, crack front require more energy to propagate when its direction and plane changes. Thus more impact energy was required when the percentage of cement was increased. The absorbed impact energy also

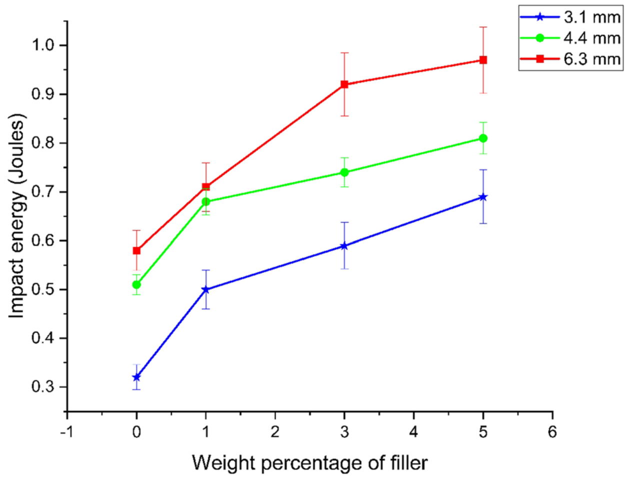

Figure 4. Variation of impact energy absorbed up to fracture with percentage of cement particles and thickness of composites.

increased with increase in thickness (as shown in figure 4) because more energy was required to fracture due to increase in the number of plies. Brittle fracture of matrix with deep cracks was observed for CmJFRP composites when the thickness was increased, as shown in figure 6.

Figure 7 shows the Southwell plot for JFRP composite (3.1 mm thickness) in case of buckling, with both ends hinged. Southwell plot is used to determine the critical load capacity of the material. Inverse of best fit line slope gives the critical load. The Southwell plots were plotted for JFRP and CmJFRP composites, and critical buckling loads were determined, as shown in figure 8.

It was observed that critical buckling (Pcr) load increased with increase in thickness of composites because increase in number of plies required more energy to fracture. The composite with 3.1 mm thickness showed very less variation in critical buckling load with the increase in percentage of cement particles. However, the critical buckling load increased with the percentage of filler for 4.4 mm and 6.3 mm thick composites. Figure 9 shows the SEM image of 1% cement filled CmJFRP composite having 3.1 mm thickness.

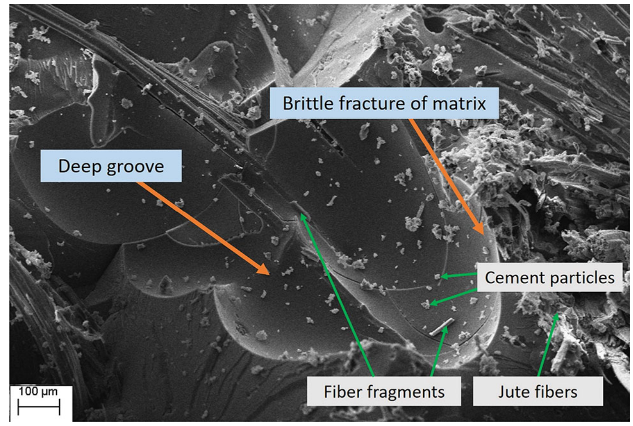

Figure 6. SEM images of fracture surfaces (Izod impact test) of 3% cement filled CmJFRP composite with 4.4 mm thickness.

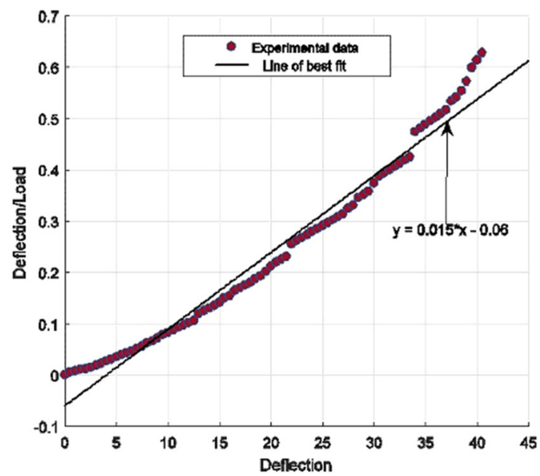

Figure 7. Southwell plot for JFRP composite (3.1 mm thickness).

It can be observed that fibers didn't deflect the crack front to a significant extent and almost failed with matrix. The cement particles were not uniformly distributed near the fiber/atrix interface and in the matrix due to less

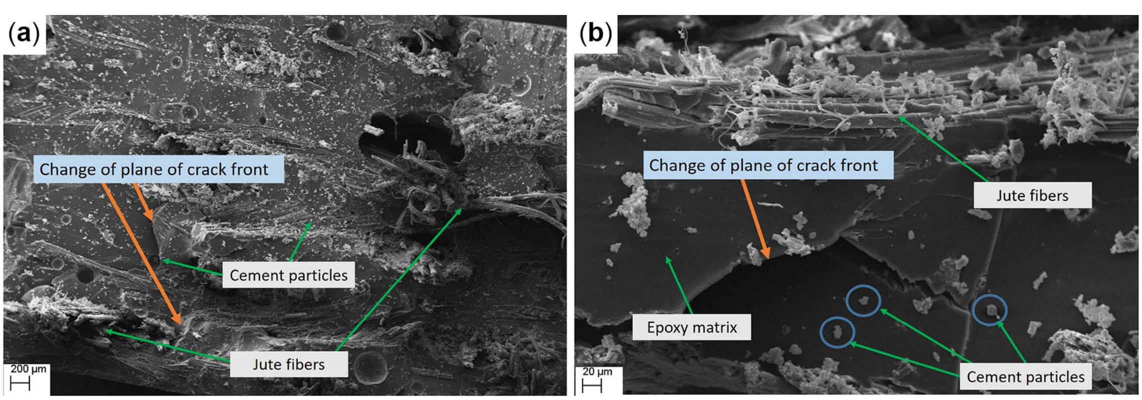

Figure 5. SEM images of fracture surfaces (Izod impact test) of 3% cement filled CmJFRP composite with 3.1 mm thickness.

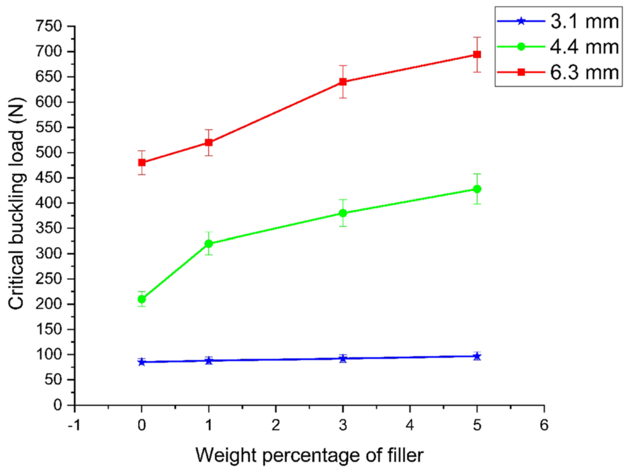

Figure 8. Variation of critical buckling load with percentage of cement particles and thickness of composites.

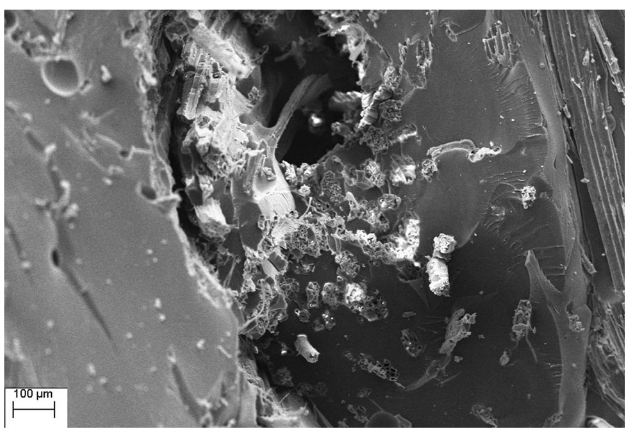

Figure 9. SEM image of 1% filled CmJFRP composite (3.1 mm thickness) failed under buckling load.

thickness. Thus, crack pinning effect by cement particles was not significant for 3.1 mm thick composites and hence, critical buckling load did not significantly vary with the percentage of filler particles.

The critical buckling load was observed to increase with increase in the percentage of filler for 4.4 mm and 6.3 mm thick composites, whereas, opposite trend was observed for tensile strength. This was attributed to the load bearing capability of cement particles in compression, whereas, the regions near cement particles acted as crack initiation and stress concentration sites when tensile stress was applied. When buckling load was applied, the crack deflection was observed in the vicinity of cement particles for 4.4 mm and 6.3 mm thick composites due to more uniform distribution of cement particles. This led to change in plane of crack propagation in the fiber filaments itself, as shown in figure 10 for 6.3 mm thick composite. The change in plane of crack required more stress level for its propagation which increased the critical buckling load with increase in percentage of cement. It can also be observed from figure 10 that matrix failure was of brittle type. Some matrix material was also absorbed by jute fibers which led to curing of matrix material at the surface and inner core of fiber filaments, and made the filaments stiffer.

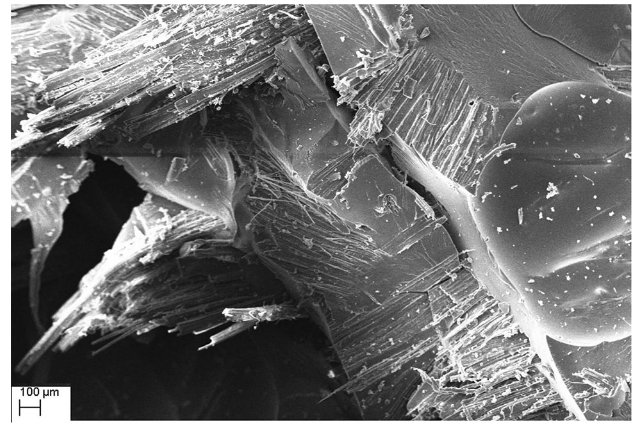

Figure 10. SEM image of 1% filled CmJFRP composite (6.3 mm thickness) failed under buckling load.

4. Conclusions

The effect of cement particles loading and thickness was investigated for jute fibers reinforced epoxy matrix composites. The effect of filler and thickness was assessed through mechanical characterization of the composites. Following conclusions were drawn from the present study.

1. Tensile strength of cement particles loaded jute fibers reinforced laminated composites decreased with increase in the percentage of filler, whereas it increased with the increase in thickness of composites.
2. The Vickers microhardness increased with increase in the percentage of filler due to presence of hard cement particles which resisted the localized plastic deformation.
3. The Izod impact tests confirmed that the impact energy absorbed upto fracture increased with increase in the percentage of filler as well as thickness.
4. The critical buckling load increased with the increase in percentage of filler and thickness of composites which was mainly ascribed to the deflection of crack front when it encountered the cement particles.

References

[1] Devireddy S B R and Biswas S 2017 Physical and mechanical behavior of unidirectional banana/jute fiber reinforced epoxy based hybrid composites. Polymer Composites 38(7): 1396–1403

[2] Srivastava V K and Kumar P. 2019 Stress field around graphene nanoplates coated carbon fiber strand reinforced in polymer matrix. *Materials Research Express*, 6(4): 045305

[3] Zhang R L, Gao B, Ma Q H, Zhang J, Cui H Z and Liu L, 2016 Directly grafting graphene oxide onto carbon fiber and the effect on the mechanical properties of carbon fiber composites. *Materials & Design*, 93: 364–369

[4] Zhou X, Dai G and Guo W 2000 Influence of functionalized polyolefin on interfacial adhesion of glass fiber-reinforced polypropylene. *Journal of Applied Polymer Science*, 76(8): 1359–1365

[5] Zhang R L, Gao B, Du W T, Zhang J, Cui H Z, Liu L, Ma Q H, Wang C G and Li F H 2016 Enhanced mechanical properties of multiscale carbon fiber/epoxy composites by fiber surface treatment with graphene oxide/polyhedral oligomeric silsesquioxane. Composites Part A: *Applied Science and Manufacturing*, 84: 455–463

[6] Li N, Wu Z, Huo L, Zong L, Guo Y, Wang J and Jian X, 2016 One-step functionalization of carbon fiber using in situ generated aromatic diazonium salts to enhance adhesion with PPBES resins. RSC Advances, 6(74): 70704–70714

[7] Alsaadi M and Erkliğ A 2017 A comparative study on mode I and mode II interlaminar behavior of borax and SiC particles toughened S-glass fabric/epoxy composite. *Arabian Journal for Science and Engineering*, 42(11): 4759–4769

[8] Gujjala R, Ojha S, Acharya S K and Pal S K 2014 Mechanical properties of woven jute-glass hybrid-reinforced epoxy composite. *Journal of Composite Materials*, 48(28): 3445–3455

[9] Saba N, Tahir P M and Jawaid M 2014 A review on potentiality of nano filler/natural fiber filled polymer hybrid composites. *Polymers*, 6(8): 2247–2273

[10] Uthayakumar M, Manikandan V, Rajini N and Jeyaraj P 2014 Influence of redmud on the mechanical, damping and chemical resistance properties of banana /polyester hybrid composites. *Materials & Design*, 64: 270–279

[11] Biswas S, Patnaik A and Kaundal R 2012 Effect of red mud and copper slag particles on physical and mechanical properties of bamboo-fiber-reinforced epoxy composites. *Advances in Mechanical Engineering*, 4: 141248

[12] Essabir H, Bensalah M O, Rodrigue D, Bouhfid R and Qaiss A 2016 Structural, mechanical and thermal properties of bio-based hybrid composites from waste coir residues: Fibers and shell particles. *Mechanics of Materials*, 93: 134–144

[13] Taib R M, Hassan H M and Mohd Ishak Z A 2014 Mechanical and morphological properties of polylactic acid/kenaf bast fiber composites toughened with an impact modifier. *Polymer-Plastics Technology and Engineering*, 53(2): 199–206

[14] Saba N, Paridah M T, Abdan K and Ibrahim N A 2016 Effect of oil palm nano filler on mechanical and morphological properties of kenaf reinforced epoxy composites. *Construction and building materials*, 123: 15–26

[15] Singh H, Singh J I P, Singh S, Dhawan V and Tiwari S K 2018 A brief review of jute fibre and its composites. *Materials Today: Proceedings*, 5(14): 28427–28437

[16] Faruk O, Bledzki A K, Fink H P and Sain M. 2012 Biocomposites reinforced with natural fibers: 2000-2010. *Progress in Polymer Science*, 37(11): 1552–1596

[17] Sharma A and Patnaik A 2018 Experimental investigation on mechanical and thermal properties of marble dust particulate-filled needle-punched nonwoven jute fiber/epoxy composite. *JOM*, 70(7): 1284–1288

[18] Dinesh S, Kumaran P, Mohanamurugan S, Vijay R, Singaravelu D L, Vinod A, Sanjay M R, Siengchin S and Bhat K S 2020 Influence of wood dust fillers on the mechanical, thermal, water absorption and biodegradation characteristics of jute fiber epoxy composites. *Journal of Polymer Research*, 27(1): 1–13

[19] Raghavendra G, Ojha S, Acharya S K, Pal S K and Ramu I 2016 Evaluation of mechanical behaviour of nanometer and micrometer fly ash particle-filled woven bidirectional jute/glass hybrid nanocomposites. *Journal of Industrial Textiles*, 45(6): 1268–1287

[20] Pawar M J, Patnaik A and Nagar R 2017 Investigation on mechanical and thermo-mechanical properties of granite powder filled treated jute fiber reinforced epoxy composite. *Polymer Composites*, 38(4): 736–748

[21] LeBow C J 2018 Effect of Cement Content on Concrete Performance

[22] Singh V, Kumar P and Srivastava V K 2018 Influence of cement particles on the mechanical and buckling behavior of laminated GFRP composites with variation of end conditions of buckling. *Materials Research Express*, 5(6): 065323

[23] Ali A, Nasir M A, Khalid MY, Nauman S, Shaker K, Khushnood S, Altaf K, Zeeshan M and Hussain A 2019 Experimental and numerical characterization of mechanical properties of carbon/jute fabric reinforced epoxy hybrid composites. *Journal of Mechanical Science and Technology*, 33(9): 4217–4226

[24] Khan G A, Terano M, Gafur M A and Alam M S 2016 Studies on the mechanical properties of woven jute fabric reinforced poly (l-lactic acid) composites. *Journal of King Saud University-Engineering Sciences*, 28(1): 69–74

[25] El Messiry M, Mito A B, Al-Oufy A and El-Tahan E 2014 Effect of fabric material and tightness on the mechanical properties of fabric–cement composites. *Alexandria Engineering Journal*, 53(4): 795–801

[26] Pawar M J, Patnaik A and Nagar R 2017 Experimental investigation and numerical simulation of granite powder filled polymer composites for wind turbine blade: a comparative analysis. *Polymer Composites*, 38(7): 1335–1352

[27] Fu S Y, Feng X Q, Lauke B and Mai Y W 2008 Effects of particle size, particle/matrix interface adhesion and particle loading on mechanical properties of particulate–polymer composites. *Composites Part B: Engineering*, 39(6): 933–961